11 Information handling, modelling and forward planning

Topic	What students need to learn:
11 Information handling, modelling and forward planning	11.2 - Modelling the costing of projects to achieve an optimum outcome: a) budgets – undertake financial forecasts b) planning for production – allocation of: ▪ *employees* ▪ *materials* ▪ *scale of production* c) selection of appropriate tools, machines and manufacturing processes.

a) Budgets – undertake financial forecasts

Budgets – undertake financial forecasts

- Accurate ***financial forecasts***, designed to gauge potential income and expenditure, are a significant element in making decisions on the potential viability of product design and manufacture.
- Production levels, retail prices and investment potential can also be gauged.
- Several software-based systems are available to assist in carrying out financial forecasts.

Budgets – undertake financial forecasts

- In many cases both short term and long term forecasts will be carried out, with reviews taking place regularly based on sales, stock held in inventory and to an extent manufacturing activities.
- Financial forecasts can help a manufacturer ***to make decisions*** related to production levels, retail prices and also the potential for investment.

Budgets – undertake financial forecasts

- Financial forecasts are based on:
  - *Past and current project information regarding income and expenditure*
  - *Market trends*
  - *Relevant government policies*
  - *Long-term aims and objectives*
  - *Short and long-term forecasts*
  - *Sales, inventory and manufacturing reviews*

b) planning for production – allocation of:

- *employees*
- *materials*
- *scale of production*

Planning for Production

- Costings will be required for:
  - *Employees*
  - *Materials*
  - *Scale of production*
- This is generally carried out using ***'Master Production Schedule' (MPS)*** – see chapter 8.3

Typical Mistake: Don’t exclude wider factors from an explanations of the importance of financial forecasts.

Government policy in terms of vehicle emissions, for example, has an enormous effect on the potential sales of proposed car designs.

c) selection of appropriate tools, machines and manufacturing processes.

Selection of Appropriate Tools, Machines and Manufacturing Processes

- Tools, machines and manufacturing processes are prerequisites for the production of products (see chapter 3.1)
- The choice of materials, its properties and costs, as well as the scale of production concerned, are the key factors which dictate the most appropriate manufacturing process.
- Here are some examples of production methods selected for particular circumstances.

Selection of Appropriate Tools, Machines and Manufacturing Processes

Here are some contrasting examples from the car industry which link manufacturing methods, materials and other related factors.

A: Economy cars (low carbon steel)

- *Highly automated presses cut and form the panels.*
- *Parts are easily fused together, so robotic arms can be programmed to spot weld them.*
- *Minimal human intervention is required.*
- *Press dies and robot arm programming can be easily and quickly changed.*
- *This facilitates the incorporation of Just In Time (JiT) manufacture to maximise profitability.*

Selection of Appropriate Tools, Machines and Manufacturing Processes

B: Prestige cars (aluminium)

- *Aluminium is increasing used, despite it being more expensive than low carbon steel, as it is lighter and more corrosion resistant.*
- *The improved power-to-weight ratio improves the car's performance and economy.*
- *It is difficult to weld aluminium, so a range of mechanical fastenings such as rivets are being used in conjunction with adhesives such as epoxy resin.*
- *The level of automation is reduced since these are more labour-intensive manufacturing methods.*

Selection of Appropriate Tools, Machines and Manufacturing Processes

C: Super cars (CFRP – Carbon Fibre Reinforced Plastic)

- *CFRP offers very high strength-to-weight ratio to optimise the car's performance.*
- *CFRP requires high precision moulds in which layers of manually cut pre-preg (carbon fibre, pre-impregnated with epoxy resin) have to be positioned in the mould by hand before curing in an 'autoclave' (oven).*
- *Automation in CFRP manufacture is developing, so it may eventually become more common in general car manufacture.*

Research Task

Research your own examples of reasons for specific manufacturing methods being used for particular products.

You will find it easier to remember if you choose products related to your own interests.

For example:

Tennis rackets, football boots, gaming models, bicycles and fashion items.

Now Test Yourself

1. *Explain, using an appropriate product example, why a study of market trends would be an important element in a financial forecast.*
2. *Select and name a product that is made by die casting and give three reasons for selecting die casting rather than sand casting.*
3. *Explain why car bodies for top selling models such as the Ford Fiesta are not made using mechanical fasteners and adhesives.*